

Muralidhar Kolimali

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Professional Summary

Recent Master's graduate in Data Science with professional experience in AI-powered applications, backend engineering, data analytics, and enterprise-scale data processing. Skilled in Python, SQL, FastAPI, machine learning, Retrieval-Augmented Generation (RAG), REST APIs, and backend automation. Experienced in developing scalable AI systems using OpenAI GPT models, PostgreSQL, vector databases, and conversational AI workflows. Strong background in data pipelines, analytics, backend development, and production support across AI and enterprise applications.

Technical Skills

Programming Languages	Python, SQL, PL/SQL, Pro*C, Shell Scripting, TypeScript/JavaScript (Basics)
AI & Machine Learning	Machine Learning, NLP, Retrieval-Augmented Generation (RAG), Prompt Engineering, Scikit-learn, YOLOv3, TensorFlow (Basics), Q-Learning, OpenAI GPT-4o-mini
Backend & APIs	FastAPI, REST APIs, Webhooks, Twilio API, API Integrations, Conversational AI Systems
Databases & Data Engineering	PostgreSQL, SQL Query Optimization, ETL Concepts, Data Pipelines, Data Validation, Analytical Functions
Cloud & DevOps	Docker, Docker Compose, Git/GitHub, CI/CD Concepts, Cloud-Native Deployment Concepts, AWS Familiarity
Data Science & Analytics	Pandas, NumPy, OpenCV, Power BI, Data Analytics, Reporting, Operational Dashboards
Vector Databases & AI Tools	FAISS, Sentence Transformers, LangChain Basics
Operating Systems & Tools	UNIX/Linux, Google Colab, VS Code
Certifications	NVIDIA LLM & Chatbot Development Workshop, Prompt Engineering Workshops

Professional Experience

Data Analytics & Data Engineering Intern 2025 – Present
Eastern Metal Works (EMW), Connecticut, USA

- Worked on operational analytics and dashboard development projects involving ETL workflows, Power BI reporting, and business data integration.
- Assisted in integrating and analyzing enterprise operational datasets to support reporting and business decision-making workflows.
- Supported backend data preparation, transformation, reporting automation, and operational analytics processes.
- Collaborated on dashboard enhancements, filtering systems, and operational reporting workflows for enterprise business use cases.

Software Engineer June 2022 – July 2024
Virtusa Consulting Services (Client: Citi India), Chennai, India

- Worked on enterprise banking migration projects involving large-scale retail banking datasets and backend operational systems.
- Developed and optimized PL/SQL queries using indexing, analytical functions, optimizer hints, and SQL tuning techniques to improve data processing efficiency.
- Automated backend operational workflows using Pro*C and Shell scripting, improving process reliability and reducing manual operational effort.
- Developed SQL-based data processing and reporting pipelines using GROUP functions, SET operators, and analytical functions.
- Supported UNIX-based production systems, backend monitoring, procedural testing, and root-cause analysis activities.
- Collaborated with cross-functional teams for backend integration, enterprise support, and large-scale data validation workflows.

Projects

AI Voice Assistant for Government Services

- Developed an AI-powered conversational voice assistant using FastAPI, OpenAI GPT-4o-mini, Twilio, PostgreSQL, and Retrieval-Augmented Generation (RAG).
- Built intelligent conversational workflows for healthcare, transportation, caregiving, and support service applications.
- Designed scalable REST APIs and backend services to process real-time voice interactions and contextual AI responses.
- Implemented vector search pipelines using FAISS and Sentence Transformers for intelligent document retrieval.
- Containerized backend services using Docker and Docker Compose for scalable deployment workflows.
- Developed backend pipelines for storing user interactions, call logs, and transcripts using PostgreSQL.

Vehicle Detection using YOLOv3

- Developed a computer vision pipeline for vehicle detection and classification using YOLOv3 and OpenCV.
- Implemented frame-by-frame object detection workflows and optimized inference handling for traffic monitoring applications.

MindMaze – Intelligent Puzzle Game

- Built an adaptive puzzle game using machine learning-driven difficulty adjustments and Q-Learning concepts.
- Developed a real-time hint recommendation system to improve user interaction and gameplay experience.

MedInSight-RAG – Biomedical AI Assistant

- Developed a Retrieval-Augmented Generation (RAG) system for biomedical question answering using BioBERT embeddings and FAISS vector search.
- Evaluated multiple LLMs including BioGPT, Flan-T5, and Phi-2 for semantic similarity, factual accuracy, and hallucination reduction.
- Built AI workflows focused on reliable biomedical information retrieval and context-aware response generation.

Workshops & Certifications

- NVIDIA LLM & Chatbot Development Workshop (2025) – Hands-on training in prompt engineering, conversational AI, LLM fine-tuning, and AI deployment workflows.
- Participated in advanced university-hosted workshops focused on scalable AI application development and model deployment concepts.

Education

Master of Science in Data Science
University of New Haven, Connecticut

May 2026
GPA: 3.5

Bachelor of Technology in Electronics and Communication Engineering
Raghu Institute of Technology, India

GPA: 8.32